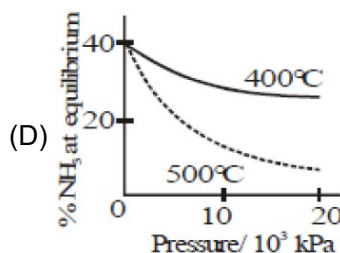
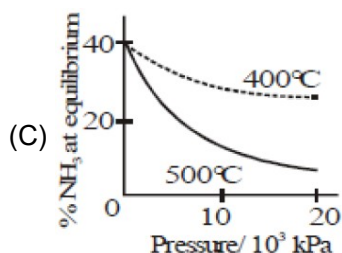
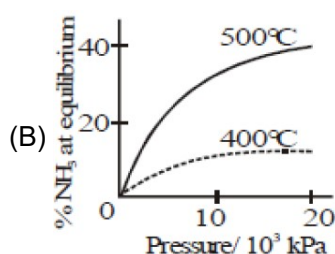
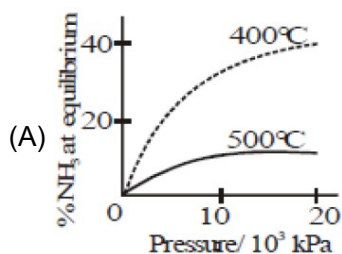


CHEMISTRY

TARGET : JEE- Advanced

CAPS-5**CHEMICAL EQUILIBRIUM
& SURFACE CHEMISTRY****SINGLE CHOICE QUESTIONS**

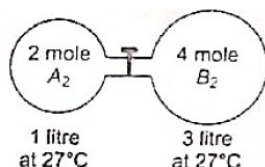
1. The percentage of ammonia obtainable, if equilibrium were to be established during the Haber process, is plotted against the operating pressure for two temperatures 400°C and 500°C. Which of the following correctly represents the two graphs?



2. The value of K_p for the reaction at 270C $\text{Br}_2(\ell) + \text{Cl}_2(\text{g}) \rightleftharpoons 2\text{BrCl}(\text{g})$ is '1 atm'. At equilibrium in a closed container partial pressure of BrCl gas is 0.1 atm and at this temperature the vapour pressure of $\text{Br}_2(\ell)$ is also 0.1 atm. Then what will be minimum moles of $\text{Br}_2(\ell)$ to be added to 1 mole of Cl_2 , initially, to get above equilibrium situation?

- (A) $\frac{10}{6}$ moles (B) $\frac{5}{6}$ moles (C) $\frac{15}{6}$ moles (D) 2 moles

3.



The gas A_2 in the left flask allowed to react with gas B_2 present in right flask as $\text{A}_2(\text{g}) + \text{B}_2(\text{g}) \rightleftharpoons 2\text{AB}(\text{g})$; $K_c = 4$ at 27°C.

What is the concentration of AB when equilibrium is established?

- (A) 1.33 M (B) 2.66 M (C) 0.66 M (D) 0.33 M



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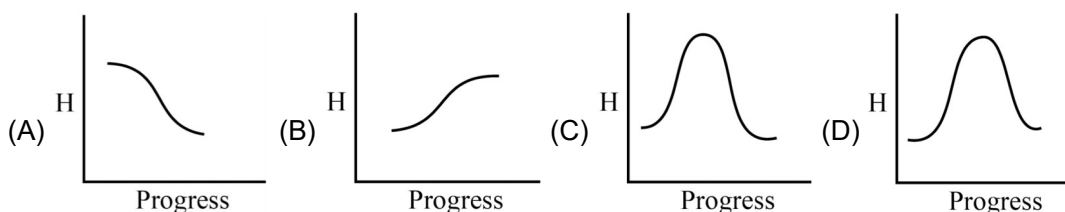
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4. For the reaction $2A(g) + B(g) \rightleftharpoons C(g) + D(g)$; $K_c = 10^{12}$. If the initial moles of A, B, C and D are 2, 1, 7 and 3 moles respectively in a one litre vessel, what is the equilibrium concentration of A?
 (A) 4×10^{-4} (B) 2×10^{-4} (C) 10^{-4} (D) 8×10^{-4}
5. One mole of SO_3 was placed in a one litre vessel at a certain temperature. The following equilibrium was established in the vessel
 $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$
 The equilibrium mixture reacts with 0.2 mole $KMnO_4$ in acidic medium. Hence, K_c is:
 (A) 0.50 (B) 0.25 (C) 0.125 (D) None of these
6. Equilibrium constants of T_2O (T is 3_1H , an isotope of 1_1H) and H_2O are different at 298 K. Let at 298 K pure T_2O has pT (like pH) is 7.62. The pT of a solution prepared by adding 10 mL of 0.2 M TCl to 15 mL of 0.25 M NaOT is:
 (A) $2 - \log 7$ (B) $14 + \log 7$ (C) $13.24 - \log 7$ (D) $13.24 + \log 7$
7. Identify the appropriate graph between enthalpy and progress of physical adsorption.

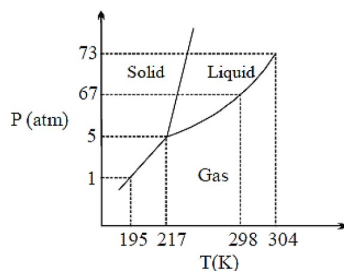


8. Which of the following is correct about lyophilic sols?
 (A) They are irreversible.
 (B) They are formed by inorganic substances.
 (C) They are readily coagulated by addition of electrolytes.
 (D) They are self-stabilized.
9. Below critical micelle concentration (CMC)
 (A) The surfactant molecules of ions undergo association to form clusters.
 (B) The viscosity of solution increases abruptly.
 (C) Substances like grease, fat, etc., dissolve colloiddally.
 (D) Salt behaves as normal, strong electrolyte.
10. A detergent ($C_{12}H_{25}SO_4Na^+$) solution becomes a colloidal sol at a concentration of 10^{-3} M. On an average 10^{13} colloidal particles are present in 1 mm³. What is the average number of ions which are contained by one colloidal particle (micelle)?
 [Given: $N_A = 6 \times 10^{23}$]
 (A) 6×10^7 (B) 10 (C) 60 (D) None of these

MULTIPLE CHOICE QUESTIONS

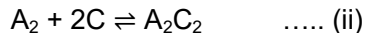
11. A positive catalyst
 (A) increases the average kinetic energy of reacting molecules
 (B) decreases the activation energy
 (C) can alter the reaction mechanism
 (D) can change pre-exponential factor
12. Which of the following is correct about the chemical equilibrium?
 (A) $(\Delta G)_{T,p} = 0$
 (B) Equilibrium constant is independent of initial concentration of reactants
 (C) Catalyst has no effect on equilibrium state
 (D) Reaction stops at equilibrium
13. $N_2(g)$ and $H_2(g)$ are allowed to react in a closed vessel at given temp. and pressure for the formation of $NH_3(g)$ [$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) + 22.4 \text{ kcal}$]. If $He(g)$ is added at equilibrium at constant pressure, then which is/are correct?
 (A) Concentration of $N_2(g)$, $H_2(g)$ and $NH_3(g)$ decrease instantly.
 (B) Moles of $NH_3(g)$ decrease
 (C) The extent of cooling depends on amount of $He(g)$ added.
 (D) Concentration of N_2 and H_2 increases and concentration of NH_3 decreases.
14. If two gases AB_2 and B_2C are mixed the following equilibria are readily established
 $AB_2(g) + B_2C(g) \rightleftharpoons AB_3(g) + BC(g)$
 $BC(g) + B_2C(g) \rightleftharpoons B_3C_2(g)$
 If the reaction is started only with AB_2 with B_2C , then which of the following is necessarily true at equilibrium:
 (A) $[AB_3]_{eq} = [BC]_{eq}$ (B) $[AB_2]_{eq} = [B_2C]_{eq}$ (C) $[AB_3]_{eq} > [B_3C_2]_{eq}$ (D) $[AB_3]_{eq} > [BC]_{eq}$

15. Phase diagram of CO_2 is shown as following



Based on above find the correct statement(s)

- (A) 298 K is the normal boiling point of liquid CO_2
- (B) At 1 atm & 190 K CO_2 will exist as gas.
- (C) $\text{CO}_2(\text{s})$ will sublime above 195 K under normal atmospheric pressure.
- (D) Melting point & boiling point of CO_2 will increase on increasing pressure.
16. The equilibrium between, gaseous isomers A, B and C can be represented as
Reaction Equilibrium constant
- $\text{A (g)} \rightleftharpoons \text{B (g)} : K = ?$
- $\text{B (g)} \rightleftharpoons \text{C (g)} : K_2 = 0.4$
- $\text{C (g)} \rightleftharpoons \text{A (g)} : K_3 = 0.6$
- If one mole of A is taken in a closed vessel of volume 1 litre, then
- (A) $[\text{A}] + [\text{B}] + [\text{C}] = 1 \text{ M}$ at any time of the reactions
- (B) Concentration of C is 4.1 M at the attainment equilibrium in all the reactions
- (C) The value of K_1 is $\frac{1}{0.24}$
- (D) None of these
17. Which statement(s) is/are correct
- (A) A solution is prepared by addition of excess of AgNO_3 solution in KI solution. The charge likely to develop on colloidal particle is positive.
- (B) The effect of pressure on physical adsorption is high if temperature is low.
- (C) Ultracentrifugation process is used for preparation of lyophobic colloids.
- (D) Gold number is the index for extent of gold plating done.
18. Select the correct statement(s)
- (A) Physisorption is favored by low temperature
- (B) Chemisorption is favored by very high temperature because the process is endothermic
- (C) Chemisorption increases with increase in temperature owing to high activation energy
- (D) Oxygen adsorbed by charcoal can be desorbed by lowering pressure and increasing temperature.

COMPREHENSION # 1 (FOR Q. 19, 20)

A_2 and C are taken in 3 : 1 mole ratio in a closed container of a certain volume at a fixed temperature and above three equilibria are established simultaneously. K_{P_1} for the first reaction is $\frac{2}{81} \text{ atm}^{-1}$. At equilibrium partial pressure of $A_4(g)$ and $AC(g)$ are found to be $\frac{1}{2} \text{ atm}$

each and the total pressure at equilibrium is found to be $\frac{27}{4} \text{ atm}$. Then, the partial pressure of

A_2C_2 at equilibrium is

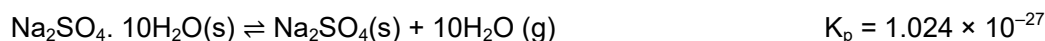
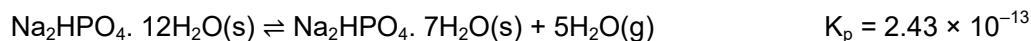
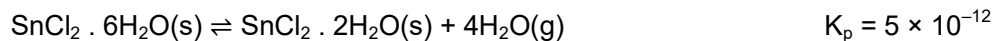
- (A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) $\frac{1}{4}$ (D) 1

20. The mole ratio of gases A_2 and AC at equilibrium is

- (A) $\frac{9}{2}$ (B) $\frac{7}{2}$ (C) 8 (D) 9

COMPREHENSION (FOR Q. 21, 22, 23)

Equilibrium constants are given (in atm) for the following reactions at 0°C :



The vapor pressure of water at 0°C is 4.56 torr.

21. Which is the most effective drying agent at 0°C ?

- (A) $\text{SnCl}_2 \cdot 2\text{H}_2\text{O}$ (B) $\text{Na}_2\text{HPO}_4 \cdot 7\text{H}_2\text{O}$ (C) Na_2SO_4 (D) all equally

22. At what relative humidity will $\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$ be efflorescent (release moisture) when exposed to air at 0°C ?

- (A) above 33.33% (B) below 33.33 % (C) above 66.66% (D) below 66.66%

23. At what relative humidity will Na_2SO_4 be deliquescent (absorb moisture) when exposed to the air at 0°C ?

- (A) above 33.33% (B) below 33.33 % (C) above 66.66% (D) below 66.66%

24. Column-I and Column-II contains four entries each. Entries of column-I are to be matched with some entries of column-II. One or more than one entries of column-I may have the matching with the same entries of column-II

Column-I

- (a) Chemisorption
(b) Physisorption
(c) Desorption of a solute on liquid surface
(d) Adsorption of a solute surface
liquid liquid

(A) $a \rightarrow Q$; $b \rightarrow P$; $c \rightarrow R$; $d \rightarrow S$

(C) $a \rightarrow P$; $b \rightarrow Q$; $c \rightarrow S$; $d \rightarrow R$

Column-II

- (P) Not specific and decreases with temperature
(Q) Specific and increases with temperature
(R) Increases the surface tension of the liquid
(S) Decreases the surface tension of the on a liquid liquid

(B) $a \rightarrow P$; $b \rightarrow Q$; $c \rightarrow R$; $d \rightarrow S$

(D) $a \rightarrow Q$; $b \rightarrow P$; $c \rightarrow S$; $d \rightarrow R$

SUBJECTIVE QUESTIONS

25. In a vessel, two equilibrium are simultaneously established at the same temperature as follows:
- $$\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g}) \quad \dots\dots(1)$$
- $$\text{N}_2(\text{g}) + 2\text{H}_2(\text{g}) \rightleftharpoons \text{N}_2\text{H}_4(\text{g}) \quad \dots\dots(2)$$
- Initially the vessel contains N_2 and H_2 in the molar ratio of 9 : 13. The equilibrium pressure is $7P_0$, in which pressure due to ammonia is P_0 and due to hydrogen is $2P_0$. Find the values of equilibrium constants (K_P 's) for both the reactions
26. 0.15 mole of CO taken in a 2.5 litre flask is maintained at 750 K along with a catalyst so that the following reaction can take place:
- $$\text{CO}(\text{g}) + 2\text{H}_2(\text{g}) \rightleftharpoons \text{CH}_3\text{OH}(\text{g})$$
- Hydrogen is introduced until the total pressure of the system is 8.5 atm at equilibrium and 0.08 mole of methanol is formed. Calculate :
- (i) K_p & K_c
- (ii) the final pressure if the same amount of CO and H_2 as before are used, but with no catalyst so that the reaction does not take place.
27. A saturated solution of iodine in water contains 0.33 g I_2/L . More than this can dissolve in a KI solution because of the equilibrium:
- $$\text{I}_2(\text{aq}) + \text{I}^-(\text{aq}) \rightleftharpoons \text{I}_3^-(\text{aq})$$
- A 0.10 M KI solution (0.10 M I^-) dissolves 12.5 g of iodine/L, most of which is converted to I_3^- . Assuming that the concentration of I_2 in all saturated solutions is the same, calculate the equilibrium constant for the above reaction. What is the effect of adding water to a clear saturated of I_2 in the KI solution?
28. Which out of 'He' and 'Ne' would adsorb on the surface of charcoal more readily and why?

ANSWER KEY OF CAPS-5

- | | | | | |
|----------|-----------|-----------|----------------------------------|----------|
| 1. (A) | 2. (C) | 3. (C) | 4. (A) | 5. (B) |
| 6. (D) | 7. (A) | 8. (D) | 9. (D) | 10. (C) |
| 11. (BC) | 12. (ABC) | 13. (ABC) | 14. (CD) | 15. (CD) |
| 16. (AC) | 17. (AB) | 18. (ACD) | 19. (C) | 20. (D) |
| 21. (A) | 22. (B) | 23. (A) | 24. (A → Q; B → P; C → R; D → S) | |
25. $(K_{p_1} = \frac{1}{20P_0^2}, K_{p_2} = \frac{3}{20P_0^2})$
26. ((i) $k_C = 189.574 \text{ mol}^{-2} \text{ lit}^2$, $k_P = 0.05 \text{ atm}^{-2}$ (ii) $P = 12.4408 \text{ atm}$)
27. (708.3 M^{-1}) 28. (Ne)

